

# Research Proposal & Motivation: On Axiomatic Approach to Quantum Field Theory

Roshan Raj

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## 1 Introduction

Quantum field theory is one of the most successful theories of physics, supported by a range of experimental verification and crucial predictions. From Qunatum Electrodynamics to The Standard Model, it has been a foundational stone for the same. Despite such acceptance and accuracy, it lacks a profound mathematical core, the presence of which could enable binding the whole theory under a unified and consistent mathematical rule. Multiple attempts have been made to identify a set of axioms based on which complete quantum field theory can be seen as a natural outcome.

This Research proposal is organized as follows: Section 2 provides a general overview of the developmental quantum field theory in lucid terms. In Section 3, a synopsized research problem is presented, with the inclusion of a list of attempts made to resolve the mathematical foundation of quantum field theory. Section 4 presents the outline of the proposal, methodology and a tabulated timeline. Finally, Section 5 highlights the key motivation which made me interested in this specific area of research.

## 2 Review

After criticizing the attempts made by A. Sommerfeld to consider unobservables in quantum theory rather than employing correct sets of observables, W. Heisenberg originally conceived quantum mechanics in his famous 1925 paper[1]. His take on observables was inspired by the special theory of relativity approach in identifying the correct sets of observables for developing theories. The work was then popularised by M. Born, P. Jordan and Heisenberg and theoretically tested by W. Pauli [2][3]. An independent formalism, reportedly inspired by L. de Broglie's consideration, was later formulated in famous notes by E. Schrödinger in 1926 [4]. P.A.M. Dirac found that both approaches are equivalent, motivated by a question raised by Schrödinger. Matrix mechanics treated quantum mechanical operators as time-dependent, morphologically similar to the general form of Hamilton's equation, and Schrödinger considered quantum states as time-dependent, which is seemingly based upon the Hamilton-Jacobi equation of classical physics [5]. Their basic constructions are respectively involving operator algebra and functional analysis, which would also later serve as a base for the algebraic and the functorial approach to quantum field theory.

The Schrödinger equation is theoretically incapable of describing photons and stands as non-relativistic formalism, as it treats time as a scalar quantity while promoting spatial coordinates to linear operators, which would have later caused major problems in quantum physics. In 1927, J. von Neumann provided the axiomatic formalism of quantum mechanics, strengthening it with the consistent mathematical framing inspired by Dirac's attempts at transformation theory [6].

The arrival of problems such as the violation of particle numbers [7] and other inconsistencies or paradoxes within relativistic quantum mechanics motivated physicists to reconcile special relativity and quantum mechanics freshly under the name of quantum field theory (QFT): a successful field theoretic approach to relativistic quantum mechanics also cumulating classical field theory, where, position and time coordinates were then considered mere labels of a more fundamental field operator. The first phase of its development is marked as the canonical quantization of the classical fields developed by Born, Heisenberg, and Jordan of the free electromagnetic field; more substantial progress was seen through Dirac's attempts to involve the quantization of the radiation field via interaction Hamiltonian terms, whose first-order perturbation explained the phenomenon of spontaneous emission and the second-order perturbation was able to account for the scattering of photons and non-relativistic Compton scattering [8]. Later, a motivation is that Photons are quantized excitations of the electromagnetic field. In 1928 and 1930, Jordan, E. Wigner, Heisenberg, Pauli, and E. Fermi showed that elementary particles are the quanta of quantum fields. However, this approach was found to have several constraints and intrinsic issues and waiting to get cured through renormalization procedures by E. Stueckelberg [9] and others since the application of higher-order perturbation theory caused problematic infinities in calculations, as was pointed out by R. Oppenheimer, such as the electron self-energy. A similar procedure of renormalization was reported by J. Schwinger and independently by R. Feynman, F. Dyson, and S. Tomonaga in the 1950s, postulating that the values of certain physical quantities (such as the mass and charge of an electron) must be equal to the values observed in experiments, which undoubtedly sounds to me similar as if Heisenberg's insights on observables were seemingly inspiring it. This allowed us to explain the electron's anomalous magnetic moment and vacuum polarisation [10]. Later, N. Bogolyubov and K. Wilson provided deep physical insights into renormalization procedures under their breakthrough work on renormalization groups, which was independently discovered by Stueckelberg and A. Petermann in 1953 [11][12].

Meanwhile, In the fifties, R. Feynman came up with the path integral formalism of quantum field theory [13][14] (Inclusion of Lagrangian formalism of analytical mechanics) much later than the 1934

work on fully Lorentz-covariant perturbation theory for quantum fields by Stueckelberg [9]. However, Feynman’s work is more heuristic, satisfactory mathematically and intuitively well-sounded and beautifully depicted through Feynmann’s diagrams. It was again based on the pragmatic-oriented Dirac’s proposal of incorporating action to compute transition amplitude. Path-integral quantization is morphologically similar to the contact transformation in analytical mechanics and quantum mechanics and specifically finds its roots in the generating function approach in analytical mechanics. QFT provides the best description of the quantum realm and interaction through its rich mathematics, and the Standard Model is one of the examples of its application.

As Michio Masujima describes in his book [5], “There is a Hamiltonian formalism and a Lagrangian formalism for quantum mechanics, just as for analytical mechanics. The wave mechanics of Schrodinger and the matrix mechanics of Heisenberg belong to the Hamiltonian formalism, while the path integral quantization of Feynman, which originates from Dirac, belongs to the Lagrangian formalism.”

However, a few years after Path integral formalism, Haag’s theorem pointed out that QFT is weak at its core and requires better mathematical rigour [15], especially when one uses perturbation theories in addition to its other drawbacks such as non-renormalizability and limited validity of Feynmann’s diagrams. Therefore, It saw problems in perturbative quantum field theories, making QFT use heuristically as a guiding principle. However, several works were done to make the Standard Model, which brought about a renaissance in QFT and subdued most problems of perturbative quantum field theories except for a lack of mathematical rigour.

### 3 Tentative Problem

The open area of research, “axiomatic approach to QFT,” aims for mathematical rigour, formally put, “What is the particular set of Axioms which can completely express the Quantum field theory mathematically?” [16]. The axiomatic approach to QFT has pursued trial & error methodology to analyze and resolve the problem by considering different sets of axioms at different points of the era [17]. In an axiomatic approach to QFT, the emphasis has shifted somewhat from quantum fields to local observables (operators). The Wightman axioms for fields are then replaced by the Haag–Kastler axioms for the algebra of observables.

#### 3.1 Summary of attempts made to solve the problem

Quantum field theory (QFT) has been the subject of various attempts at axiomatization and mathematical foundation. The development of these approaches involved contributions from several prominent physicists and mathematicians. Here’s a brief overview of various attempts at axiomatization of QFT :

1. Dirac, Jordan, Neumann, and Wigner (1930s-1940s): Paul Dirac’s work laid the foundation for the later development of QFT. He introduced the concept of canonical quantization and the Dirac bracket. Pascual Jordan, John von Neumann, and Eugene Wigner made significant contributions to the mathematical formulation of quantum mechanics. After the axiomatization of quantum mechanics, Von Neumann’s work on operator algebras, rings of operators (von Neumann algebras), and continuous geometry remains influential [18]. However, subsequent development showed that von Neumann’s choice of factor type for the foundations of physics was incorrect.
2. Wightman Axioms (1950s-1960s): To provide a rigorous basis for Dirac’s formal framework, Schwartz’s theory of distributions (1945; 1950-1951) played a crucial role and was later developed

in the rigged Hilbert space framework. Arthur Wightman formulated the Wightman axioms in the 1950s, which provided a set of conditions for the construction of a quantum field theory [19]. Wightman’s axiomatic QFT has its roots in Schwartz’s theory of distributions and was later developed in the rigged Hilbert space framework.

3. Haag and Kastler (1950s-1960s): Res Jost, R. Haag, and Daniel Kastler developed the Haag–Kastler axioms, an early attempt at axiomatizing quantum field theory. These axioms emphasized the algebraic structure of local observables, which was slightly based on von Neumann’s operator algebra approach. It also later resulted in Algebraic QFT [20].
4. Osterwalder-Schrader Reconstruction (1970s): Osterwalder-Schrader reconstruction theorem developed in the 1970s [21]. This theorem played a crucial role in establishing the connection between Euclidean field theory and Minkowskian QFT.
5. Algebraic QFT (1980s-present): Basically, I. Gelfand and Neumark’s 1943 paper introduced abstract  $C^*$ -algebras, a fundamental class of normed rings. Gelfand and I. M. Segal made significant contributions to the development of the mathematical framework for QFT [22]. They introduced the concept of a GNS construction (Gelfand–Naimark–Segal construction) related to states on operator algebras. Algebraic quantum field theory (AQFT) is a prominent axiomatic approach to quantum field theory (QFT) that focuses on local operators and their algebraic relations [23]. AQFT has its roots in von Neumann’s work on operator algebras, which he developed to generalize the Hilbert space framework. An alternative approach to AQFT is FQFT (Functorial QFT (1990s-present), which focuses on states rather than observables [19].
6. Constructive QFT: The constructive QFT program, initiated by Res Jost and developed further by various researchers, including Gian Carlo Wick, Sergio Doplicher, Jacques Magnen, and others, aims to construct QFT rigorously, addressing issues such as the existence and uniqueness of the interacting theory [24].
7. Topological Quantum Field Theory (TQFT): Edward Witten and others contributed to the development of topological quantum field theory, a specialized form of QFT that focuses on topological invariants and is independent of the metric structure of spacetime [25].

It’s important to note that the quest for a complete and satisfactory mathematical foundation for QFT is ongoing, and various approaches continue to be explored by researchers in the field. Also, no single set of axioms has been universally accepted by the QFT community. Each of these contributions has played a role in advancing our understanding of the mathematical structures underlying quantum field theory.

## 4 Proposal to solve the problem

### 4.1 Objectives

- Given the lack of a specific research title at present. It is suggested to focus on making at least a countable little contribution to attempts made towards axiomatic approaches to quantum field theory.
- To strive to end the about ninety-year-old unsettled dilemma between pragmatic and axiomatic approaches to quantum field theory and to provide a mathematically consistent representation and physically sounding interpretation of the same.

| Plan of work            | Tentative Years Of Doctoral Research |   |   |           |    |    |            |    |    |             |    |    |            |   |   |
|-------------------------|--------------------------------------|---|---|-----------|----|----|------------|----|----|-------------|----|----|------------|---|---|
|                         | Self-prep.                           |   |   | I:2024-25 |    |    | II:2025-26 |    |    | III:2026-27 |    |    | IV:2027-28 |   |   |
|                         | -                                    | - | - | 10        | 02 | 06 | 10         | 02 | 06 | 10          | 02 | 06 | -          | - | - |
| General Reading         |                                      |   |   |           |    |    |            |    |    |             |    |    |            |   |   |
| Specific Courses        |                                      |   |   |           |    |    |            |    | ?  |             |    |    |            |   |   |
| Review                  |                                      |   |   |           |    |    |            |    |    |             |    |    |            |   |   |
| Specific Research Title |                                      |   |   | ?         |    | ?  |            |    |    |             |    |    |            |   |   |
| Discussion              |                                      | ? |   |           |    |    |            |    |    |             |    |    |            |   |   |
| Literture Review        |                                      |   |   |           |    |    |            |    |    |             |    |    |            |   |   |
| Drafting                |                                      |   |   |           |    |    |            |    |    |             |    |    |            |   |   |
| Presentation            |                                      |   |   |           |    |    |            |    | ?  |             |    |    |            |   |   |
| Manuscript Writing      |                                      |   |   |           |    |    |            |    |    |             |    |    | ?          |   |   |
| Presentation, Viva Voce |                                      |   |   |           |    |    |            |    |    |             |    |    | ?          | ? | ? |

Table 1: Working schedule with tentative dates for completing PhD dissertation, if granted four years long option. Shaded areas indicate the active period, and ‘?’ depicts the ‘open-ended’ duration.

- To present a detailed review of attempts made in axiomatic approach to quantum field theory while highlighting existing similarities and differences present among them. That may enable us to encircle the cause of noises that previous attempts were made, and it may provide the rationale behind the ineffectiveness of those attempts and the reason for such long-standing dilemmas.

## 4.2 Methodology

1. Given a mathematically consistent or infinitesimal free formalism of classical field theory, which can apparently reflect certain general features that an axiomatic field theory should need to carry, though classically. Then, pluck out a set of axioms embedded into such classical theory such that it can necessarily help in guessing a similar set of axioms for QFT.
2. If needed, opting for a shift in the definition, physical understanding and representation of the observable based on the clear distinctiveness of quantum fields theory and quantum mechanics.
3. Execute steps 1 and 2 to construct the relativistically & quantum physically consistent set of axioms for quantum field theory.

## 4.3 Timeline

The following working schedule (see Table 1) aims to present the tentative deadlines for the completion of consequent takes, as one expects to encounter.

## 5 Motivation

Broadly, I am highly motivated to research and earn my PhD in Theoretical Physics by writing my PhD thesis on the axiomatic approach to QFT.

My immediate motivation towards the axiomatic approach to QFT arose after the following realizations:

- It is a natural extension of the quantum foundation, which I have been extensively studying and working on since my third year in university, as reflected by my last few UG research projects and the master’s dissertation.

- J. von Neumann’s work on the mathematical foundation of Quantum mechanics can be viewed as the first mathematical rigour of the subject. It seems that we need a second mathematical rigour through identifying the correct and consistent set of axioms for QFT in general.
- The axiomatic approach seeks to provide a rigorous mathematical foundation for QFT and thus provide a better and broader view. It also serves much freedom of research rather than learning about ordinary QFT and then calculating transition amplitude and doing all sorts of applied works. Thus, I count axiomatic approaches to QFT above ordinary QFT since it is supportive of my research temperament and makes me feel open-minded as a seeker, not a believer.
- My timeline of getting introduced to Lagrangian QFT seems to highlight that I have been moving towards the subject of mathematical understanding of QFT. Lectures by A. Alekseev, D. Tong, and P. Mukhopadhyay, and works of literature by M. Sredniki, D. Tong, J. Conlon, P. van Baal, Notes by G. ’t Hooft, S. Chatterjee, A. Jaffe since eight months. Out of these, my immediate motivation to opt for the axiomatic QFT came from going through [26],[27],[28],[19] short lectures and notes.

## 6 Declaration

I, Roshan Raj, hereby declare that the research proposal draft titled *On Axiomatic Approach to Quantum Field Theory* is my effort and has not been copied or plagiarized in any way. I have cited all sources used in the proposal. I would like to formally declare that an AI text tool was employed to prepare an outline of sub-section 3.1 to enhance grammar and address other aspects. However, I have cited all sources, including the online generator, in the proposal.

I believe that it is a well-conceived and feasible plan for research. I am committed to conducting my research in an ethical and responsible manner.

## References

- [1] W. Heisenberg. A quantum-theoretical reinterpretation of kinematic and mechanical relations. *Z. Phys.*, 33:879–893, 1925.
- [2] M. Born, W. Heisenberg, and P. Jordan. Zur Quantenmechanik. II. *Z. Phys.*, 35(8-9):557–615, 1926.
- [3] W. Pauli. Zur Quantenmechanik des magnetischen Elektrons. *Zeitschrift für Physik*, 43(9-10):601–623, 1927.
- [4] E. Schrödinger. Quantisierung als Eigenwertproblem. *Annalen Phys.*, 384(4):361–376, 1926.
- [5] Michio Masujimo. *Path Integral Quantization and Stochastic Quantization*. Springer-Verlag, 2nd edition, 2009.
- [6] John von Neumann and Robert Beyer. *Mathematical Foundations of Quantum Mechanics: New Edition*. Princeton University Press, 2018.
- [7] D.A. Bromley and W. Greiner. *Relativistic Quantum Mechanics. Wave Equations*. Springer Berlin Heidelberg, 2013.

- [8] Paul A. M. Dirac. Quantum theory of emission and absorption of radiation. *Proc. Roy. Soc. Lond. A*, 114:243, 1927.
- [9] J. Lacki, H. Ruegg, and G. Wanders. *E.C.G. Stueckelberg, An Unconventional Figure of Twentieth Century Physics: Selected Scientific Papers with Commentaries*. Birkhäuser Basel, 2009.
- [10] V. F. Weisskopf. The development of field theory in the last 50-years. *Phys. Today*, 34N11:69–85, 1981.
- [11] K. G. Wilson and John B. Kogut. The Renormalization group and the epsilon expansion. *Phys. Rept.*, 12:75–199, 1974.
- [12] Ernst Carl Gerlach Stueckelberg de Breidenbach and Andreas Petermann. Normalization of constants in the quanta theory. *Helv. Phys. Acta*, 26:499–520, 1953.
- [13] Richard Phillips Feynman. *The principle of least action in quantum mechanics*. PhD thesis, Princeton U., 1942.
- [14] R. P. Feynman. Space-time approach to nonrelativistic quantum mechanics. *Rev. Mod. Phys.*, 20:367–387, 1948.
- [15] R. Haag. On quantum field theories. *Kong. Dan. Vid. Sel. Mat. Fys. Med.*, 29N12:1–37, 1955.
- [16] Arthur Jaffe and Edward Witten. Quantum yang-mills theory, 2000.
- [17] Detlev Buchholz. Current trends in axiomatic quantum field theory, 1998.
- [18] Pascual Jordan, J. von Neumann, and Eugene P. Wigner. On an Algebraic generalization of the quantum mechanical formalism. *Annals Math.*, 35:29–64, 1934.
- [19] Fred Kronz and Tracy Lupher. Quantum Theory and Mathematical Rigor. In Edward N. Zalta, editor, *The Stanford Encyclopedia of Philosophy*. MRL, Stanford University, Winter 2021 edition, 2021.
- [20] R. Haag. *Local Quantum Physics: Fields, Particles, Algebras*. Theoretical and Mathematical Physics. Springer Berlin Heidelberg, 2012.
- [21] Konrad Osterwalder and Robert Schrader. Axioms for Euclidean Green’s functions II. *Communications in Mathematical Physics*, 42(3):281–305, October 1975.
- [22] M. Neumark I. Gelfand. On the imbedding of normed rings into the ring of operators in hilbert space. *Matematicheskij sbornik*, 54(2):197–217, 1943.
- [23] Christopher J. Fewster and Kasia Rejzner. Algebraic Quantum Field Theory – an introduction, 4 2019.
- [24] James Glimm and Arthur Jaffe. *Quantum Field Theory Models*, pages 11–121. Birkhäuser Boston, Boston, MA, 1985.
- [25] Edward Witten. Topological Quantum Field Theory. *Commun. Math. Phys.*, 117:353, 1988.
- [26] Arthur Jaffe. Constructive quantum field theory.
- [27] Sourav Chatterjee. Yang-mills for mathematicians.
- [28] Meinard Kuhlmann. Quantum Field Theory. In Edward N. Zalta and Uri Nodelman, editors, *The Stanford Encyclopedia of Philosophy*. MRL, Stanford University, Summer 2023 edition, 2023.